

UNIVERSITY OF MUMBAI



MASTER OF COMPUTER APPLICATION (MCA)

In-Semester Capstone Project Report

On

TOURISM MANAGEMENT SYSTEM

Submitted By: -

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Under the Guidance of

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Thakur Educational Trust's (Regd.)

THAKUR INSTITUTE OF MANAGEMENT STUDIES, CAREER DEVELOPMENT & RESEARCH

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MASTER OF COMPUTER APPLICATION (MCA)



CERTIFICATE

This is to certify that the project entitled “**Tourism Management System**” has been successfully undertaken at the **University of Mumbai** by **Vishwakarma Sumant Ramsingar (Roll No. 218)** and **Yadav Aman Kumar Rajkumar (Roll No. 225)**.

This project is submitted in **partial fulfillment of the requirements for the award of the degree of Master of Computer Application (MCA), Semester I**, during the **Academic Year 2025–2027**, and does not form a part of any other course previously undertaken by the students.

It is further certified that the students have satisfactorily completed **all the required phases of the project** under the prescribed guidelines.

Project Guide

HOD

**External Examiner
Examiner**

Internal

College Seal

Abstract

Tourism has become one of the most significant contributors to economic growth and cultural exchange across the world. With the rapid advancement of internet and web technologies, travelers increasingly prefer online platforms to search, compare, and book tour packages instead of relying on traditional travel agencies. However, many existing systems lack centralized management, transparency, and user-friendly interfaces, leading to inefficiencies and inconvenience for both tourists and service providers.

The **Tourism Management System** is a web-based application developed to address these challenges by providing an integrated and accessible platform for managing tourism services. The system enables users to register, log in, explore available tour packages, select preferred travel dates, and submit booking requests online. At the same time, administrators can efficiently manage tour packages, review booking requests, approve or reject bookings, and handle user feedback or complaints.

By digitizing the traditional tour booking process, this system reduces manual effort, improves communication between users and administrators, and enhances overall operational efficiency. It promotes transparency by allowing users to track booking status and manage their travel plans conveniently. The Tourism Management System aims to simplify trip planning, improve user experience, and provide a reliable foundation for future enhancements such as online payments, mobile application support, and intelligent tour recommendations.

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our project guide, **Prof. Shubham Mishra**, for his valuable guidance, constant support, and constructive feedback throughout the development of this project. His expertise and encouragement played a crucial role in helping us understand the project requirements clearly and successfully complete the work.

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Chapter 1

1. INTRODUCTION

1.1 Introduction of the Project

Tourism plays an important role in the economic growth of a country. With the rapid development of internet technology, people now prefer online platforms to search, compare and book tour packages instead of visiting travel agencies physically.

The **Tourism Guide & Tour Booking System** is a web-based application designed to provide users with an easy and efficient way to explore different tour packages offered by the website. The system allows users to register, log in, view tour details and book tours online.

The application consists of two main modules: **Admin Module** and **User Module**. The admin manages tour packages and approves or rejects booking requests, while users can book tours, cancel bookings and provide feedback or extra service requests.

1.2 Project Purpose

The main purpose of this project is to digitize the traditional tour booking process and provide a centralized online platform for managing tourism services.

This system aims to:

Reduce manual work involved in tour booking

Improve communication between users and the admin

Provide a transparent and user-friendly booking experience

Allow users to easily plan and manage their trips

1.3 Project Objective

The objectives of the Tourism Guide & Tour Booking System are:

To develop a secure login system for admin and users

To display different tour packages to users

To allow users to book tours by selecting preferred dates

To provide admin approval for booking confirmation

To enable users to cancel booked tours

To collect feedback and extra service requests from users

1.4 Scope of the Project

The scope of this project includes the following features:

User registration and login

Admin login and management dashboard

Tour package listing with details

Tour booking with date selection

Booking approval or rejection by admin

Tour cancellation by user

Feedback and extra service request functionality

The project is designed as a web application and can be extended in the future to include online payment, mobile application support and advanced tour recommendation features.

Chapter 2

2. SYSTEM STUDY

System study is an important phase of software development in which the current system is analysed and the requirements of the proposed system are clearly understood. This phase helps in identifying system needs, planning the project activities and selecting the appropriate platform for development. The system study for the Tourism Guide & Tour Booking System is discussed below.

2.1 Requirement Analysis

Requirement analysis is the process of understanding what the system should do and how it should perform. In this project, requirements are identified by analysing the needs of both users and the administrator.

From the user's point of view, the system should allow easy registration and login. After logging in, users should be able to view different tour packages, select a suitable tour, choose a preferred date and place a booking request. Users should also be able to cancel their booking if required and provide feedback or request extra services related to the tour.

From the administrator's point of view, the system should provide a secure login and a dashboard to manage tour packages. The admin should be able to view booking requests sent by users and decide whether to approve or reject them. The admin should also have access to user feedback and service requests.

Apart from functional requirements, the system must also satisfy non-functional requirements such as security, reliability, good performance and user-friendly design. The system should respond quickly and protect user data from unauthorized access.

2.2 Planning and Scheduling

Planning and scheduling help in completing the project in a systematic manner within the given time frame. The development of this project was divided into different stages to ensure smooth progress.

Initially, the requirements were gathered and analyzed. After that system design was carried out using diagrams and models. In the next stage, coding was performed for both admin and user modules. Once development was completed, the system was tested to identify and fix errors. Finally, documentation and project presentation were prepared. This step-by-step planning helped in completing the project efficiently and on time.

2.3 Preliminary Product Description

The Tourism Guide & Tour Booking System is a web-based application designed to provide online tour information and booking services. The system allows users to explore various tour packages offered by the website and book tours without visiting a travel agency.

The product includes features such as tour package listing, booking request generation, admin approval of bookings, booking cancellation and feedback submission. The system reduces manual

work and improves communication between users and the tour administrator, making the booking process more efficient and transparent.

2.4 Justification of Platform

The web platform was chosen for this project because it provides easy accessibility and flexibility. A web-based system allows users to access the application from any device with an internet connection, without installing additional software.

Web technology also supports faster development easy maintenance and future scalability. Since tourism services are widely used online, a web-based solution is the most suitable platform for this project. It also helps in reducing operational costs and improving overall system efficiency.

2.5 Conceptual Model

The conceptual model of the system explains how different components interact with each other. In this system, users interact with the website to view tour packages and place booking requests. These requests are stored in the database and forwarded to the admin for approval.

The admin reviews the booking requests and updates their status as approved or rejected. Once approved, the booking confirmation is sent to the user. The database stores all information related to users, tours, bookings and feedback. This conceptual flow ensures proper coordination between users, admin and the system.

Chapter 3

3. ANALYSIS AND DESIGN

The analysis and design phase is a crucial part of the software development life cycle. In this phase, the requirements gathered during the system study are converted into a structured design. This helps in defining the system architecture, data flow, database structure and overall working of the application before implementation.

3.1 Hardware Requirements

The following hardware is required to develop and run the Tourism Guide & Tour Booking System:

- Processor: Intel Core i3 or higher
- RAM: Minimum 4 GB
- Hard Disk: Minimum 100 GB free space
- Internet Connection: Required for web access

3.2 Software Requirements

The software requirements for the project are as follows:

- Operating System: Windows 10 / Linux
- Frontend Technologies: HTML, CSS, JavaScript
- Backend Technology: Java / PHP / Python (as per implementation)
- Database: MySQL
- Web Server: Apache / Tomcat
- IDE: VS Code / Eclipse / NetBeans
- Browser: Google Chrome, Mozilla Firefox

3.3 Gantt Chart



The Gantt Chart visually represents the timeline of the **Tourism Management System** project. It shows the start and end dates of each project phase, task duration and the contribution percentages of both team members (**Vishwakarma Sumant** and **Yadav Amankumar**). The chart helps in understanding how the project activities are planned and executed over a specific time period.

Key Components of the Gantt Chart

1. **Tasks:**
Each row in the Gantt chart represents a specific phase or task of the Tourism Management System project. The tasks are arranged sequentially based on their execution order.
2. **Start Date and End Date:**
These columns indicate the planned beginning and completion dates of each task, ensuring timely project execution.
3. **Duration (Days):**
This specifies the total number of days required to complete each task.

4. **Team Member Contributions:**

The percentage contribution of each team member (**Sumant** and **Aman**) is shown for every task, indicating workload distribution.

Project Phases in the Gantt Chart

1. Planning Phase

- **Task:** Planning
- **Duration:** As per Gantt Chart
- **Sumant Contribution:** 50%
- **Aman Contribution:** 50%

Details:

This initial phase includes defining project objectives, identifying system requirements, preparing the project schedule and allocating responsibilities. Both team members contributed equally during this phase.

2. System Analysis and Design

- **Task:** System Analysis and Design
- **Duration:** As per Gantt Chart
- **Sumant Contribution:** 30%
- **Aman Contribution:** 70%

Details:

This phase focuses on requirement analysis, preparation of system diagrams such as DFD, ERD, Use Case Diagram and designing the overall system architecture. Sumant played a major role in analysis and documentation, while Aman assisted in design preparation.

3. System Planning

- **Task:** System Planning
- **Duration:** As per Gantt Chart
- **Sumant Contribution:** 50%
- **Aman Contribution:** 50%

Details:

In this phase, implementation strategies were finalized, system modules were defined and resources were planned. Both members contributed equally to system planning.

4. Implementation

- **Task:** Implementation
- **Duration:** As per Gantt Chart
- **Sumant Contribution:** 70%
- **Aman Contribution:** 30%

Details:

This phase involved coding and development of the Tourism Management System using web technologies. Features such as user registration, login, tour package management, booking, admin approval and feedback modules were implemented. Aman led the development work, while Sumant provided support and integration.

5. Testing and Validation

- **Task:** Testing and Validation
- **Duration:** As per Gantt Chart
- **Sumant Contribution:** 60%
- **Aman Contribution:** 40%

Details:

The system was tested using unit testing and integration testing to ensure correct functionality. Both team members equally contributed to testing and bug fixing.

Purpose of Gantt Chart in Project Management

- **Time Management:**
Helps in completing tasks within the planned schedule.
- **Resource Allocation:**
Clearly shows how work is distributed between team members.
- **Progress Tracking:**
Enables monitoring of completed, ongoing and pending tasks.
- **Risk Management:**
Identifies possible delays and helps adjust timelines accordingly.

Color Indication for Gantt Chart

If colors are used in the Gantt Chart, they represent task status as follows:

- **Blue:** Completed tasks
- **Green:** Ongoing tasks
- **Yellow:** Upcoming tasks
- **Red:** Delayed tasks

3.4 System Design

The System Design of the Tourism Management System defines the overall architecture, components and interaction flow of the application. The system is designed as a web-based application following a client–server architecture, ensuring scalability, security and ease of access.

Use Case Diagram:

A Use Case Diagram is a UML diagram that shows how different users interact with a system to perform specific functions.

Use Case Diagram of User (Tourist)

The User (Tourist) interacts with the system to browse tour packages and manage bookings. The following use cases are performed by the user:

Register / Login

User registers and logs into the system using personal details.

View Tour Packages

User browses available tour packages with complete details.

Book Tour

User selects a tour package, chooses a preferred date and sends a booking request.

View Booking Status

User checks whether the booking is pending, approved, or rejected.

Feedback / Raise Complaint

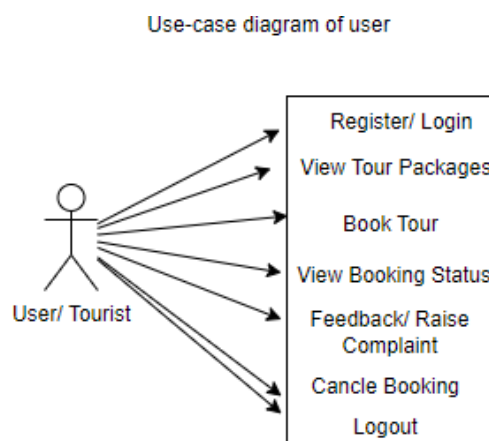
User submits feedback or raises complaints related to tours or services.

Cancel Booking

User cancels a booked tour if required.

Logout

User logs out securely from the system.



Use Case Diagram of Admin

The Admin is responsible for managing and controlling the overall functioning of the Tourism Management System. The following use cases are performed by the admin:

Register / Login

Admin logs into the system using valid credentials to access the admin dashboard.

Manage Tour Packages

Admin can add new tour packages, update existing packages, or delete tour packages.

Manage Bookings

Admin views all booking requests submitted by users.

Confirm Booking Status

Admin approves or rejects tour booking requests and updates their status.

Manage Complaints

Admin views and responds to complaints raised by users.

Add / Update / Delete Extra Places

Admin manages additional tourist places or services related to tours.

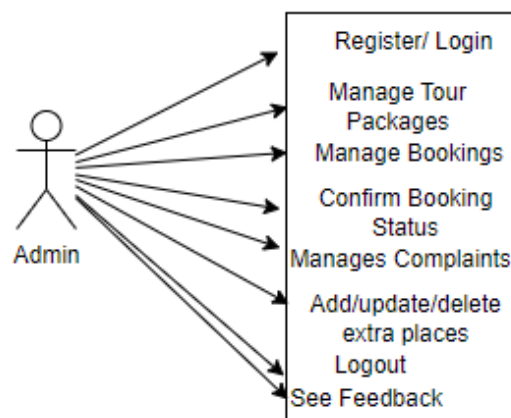
See Feedback

Admin views feedback provided by users to improve system services.

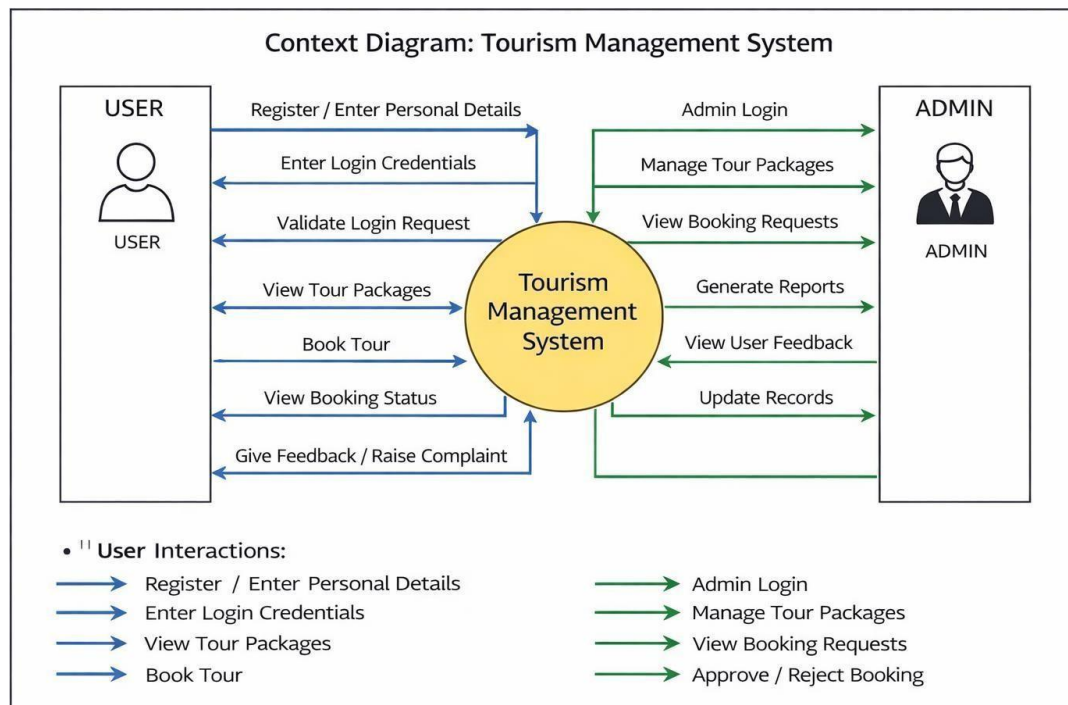
Logout

Admin securely logs out of the system.

Use-case diagram of Admin



Context Diagram:



User Interactions

- **Register / Enter Personal Details:**
New users enter personal details such as name, email ID, mobile number and password to create an account in the system.
- **Enter Login Credentials:**
Registered users log in by providing valid email ID and password to access the system.
- **Validate Login Request:**
The system verifies the user credentials and grants access if they are valid.
- **View Tour Packages:**
Users can browse available tour packages along with details such as destination, price and duration.
- **Book Tour:**
Users select a tour package, choose a preferred travel date and submit a booking request.
- **View Booking Status:**
Users can check whether their booking is pending, approved, or rejected.
- **Give Feedback / Raise Complaint:**
Users can submit feedback or raise complaints related to tour services.

- **Admin Interactions**

- **Admin Login:**
The admin logs in using secure credentials to access administrative functionalities.
- **Manage Tour Packages:**
Admin can add, update, or delete tour packages displayed to users.
- **View Booking Requests:**
Admin reviews booking requests submitted by users.
- **Approve / Reject Booking:**
Admin approves or rejects tour booking requests and updates booking status.
- **Generate Reports:**
Admin can generate reports related to bookings, users and tour packages for analysis.
- **View User Feedback:**
Admin views feedback and complaints submitted by users.
- **Update Records:**
Admin updates booking records, tour details, or user-related information when required.

- **System Role**

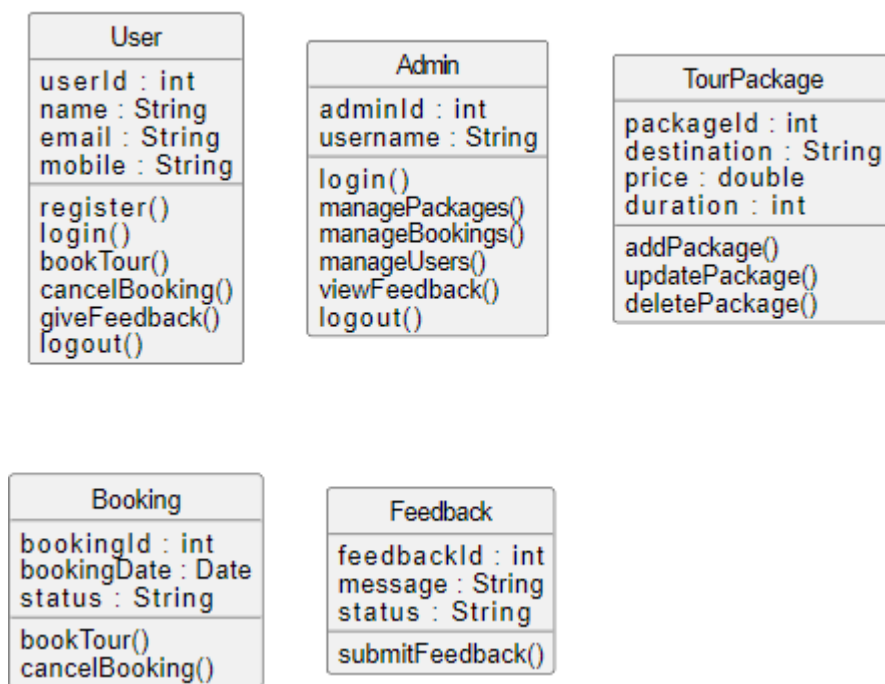
The **Tourism Management System** acts as a central platform that:

- Processes user and admin requests
- Stores and retrieves data from the database
- Ensures secure authentication
- Facilitates communication between users and administrators

Class Diagram:

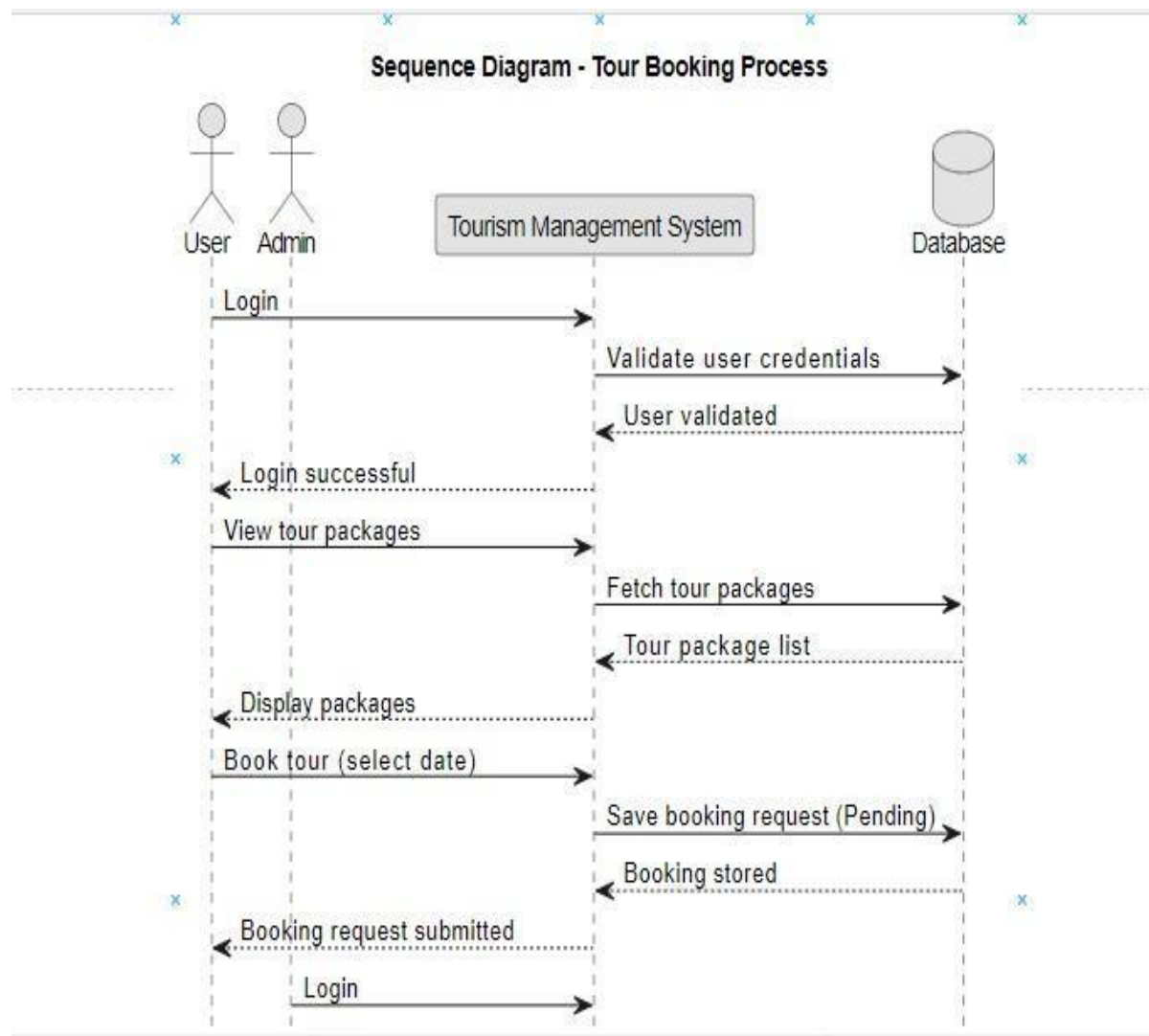
A class diagram represents the static structure of the Tourism Management System by showing its main classes, their attributes and operations. It helps in understanding how different system components such as User, Admin, Tour Package, Booking and Feedback are organized.

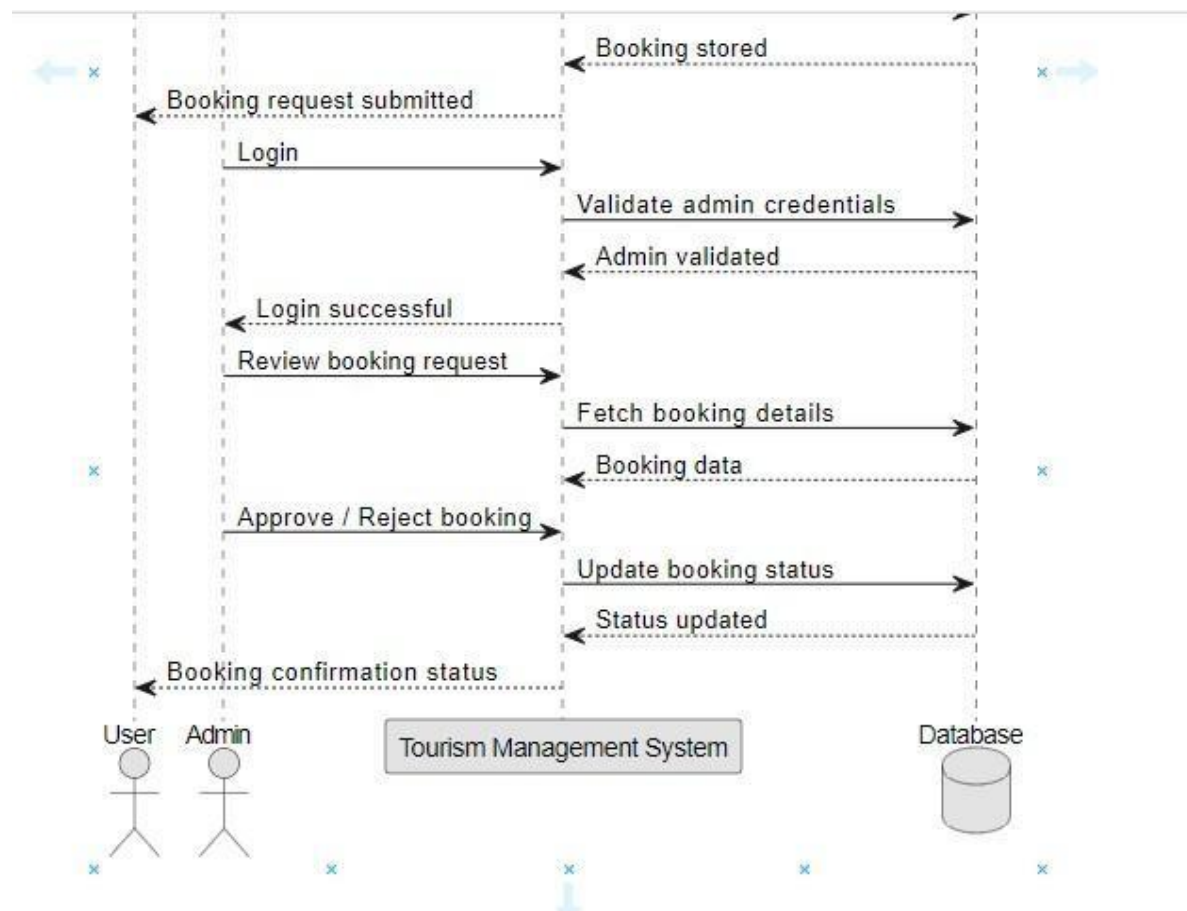
Class Diagram - Tourism Management System



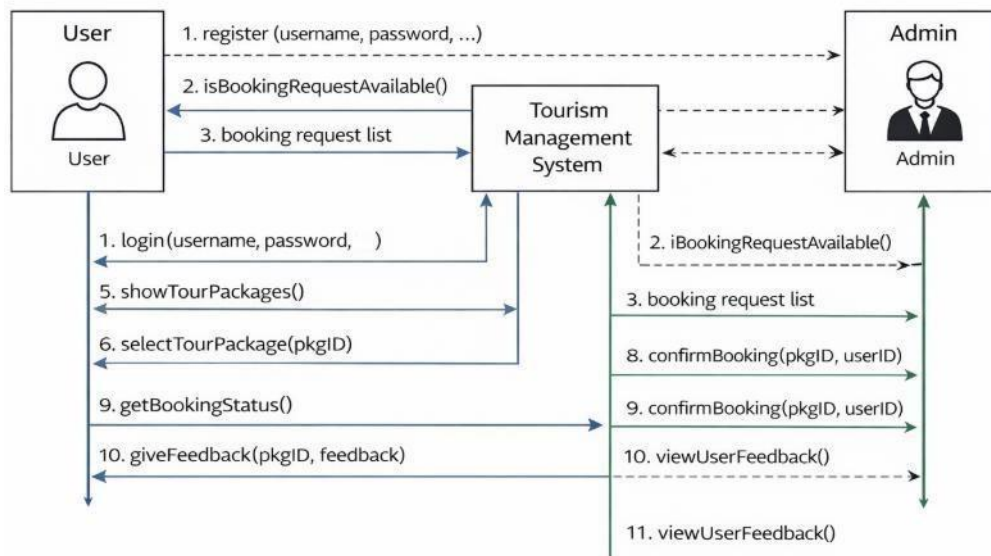
Sequence Diagram:

A sequence diagram shows the order of interaction between the user, system and admin over time. It helps in understanding the step-by-step flow of operations such as tour booking and approval in the Tourism Management System.

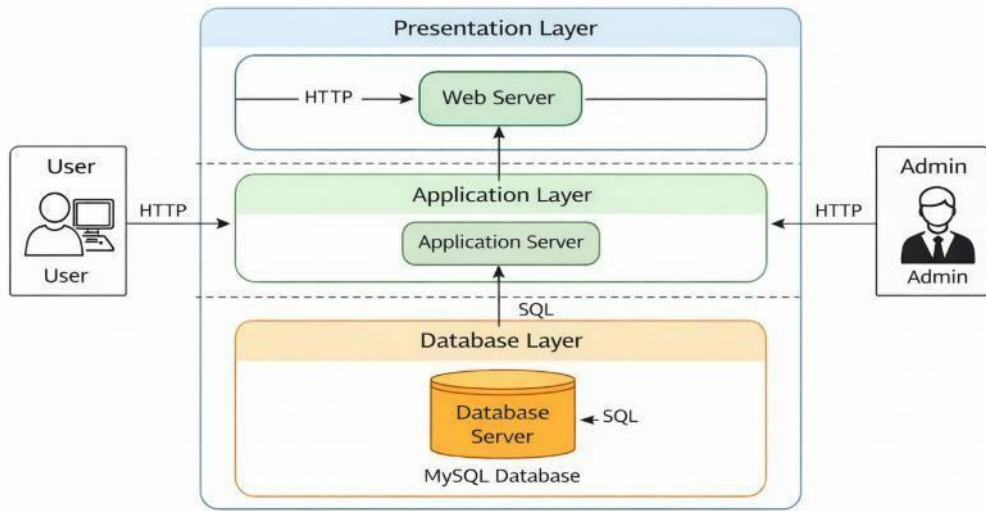




Collaboration Diagram:



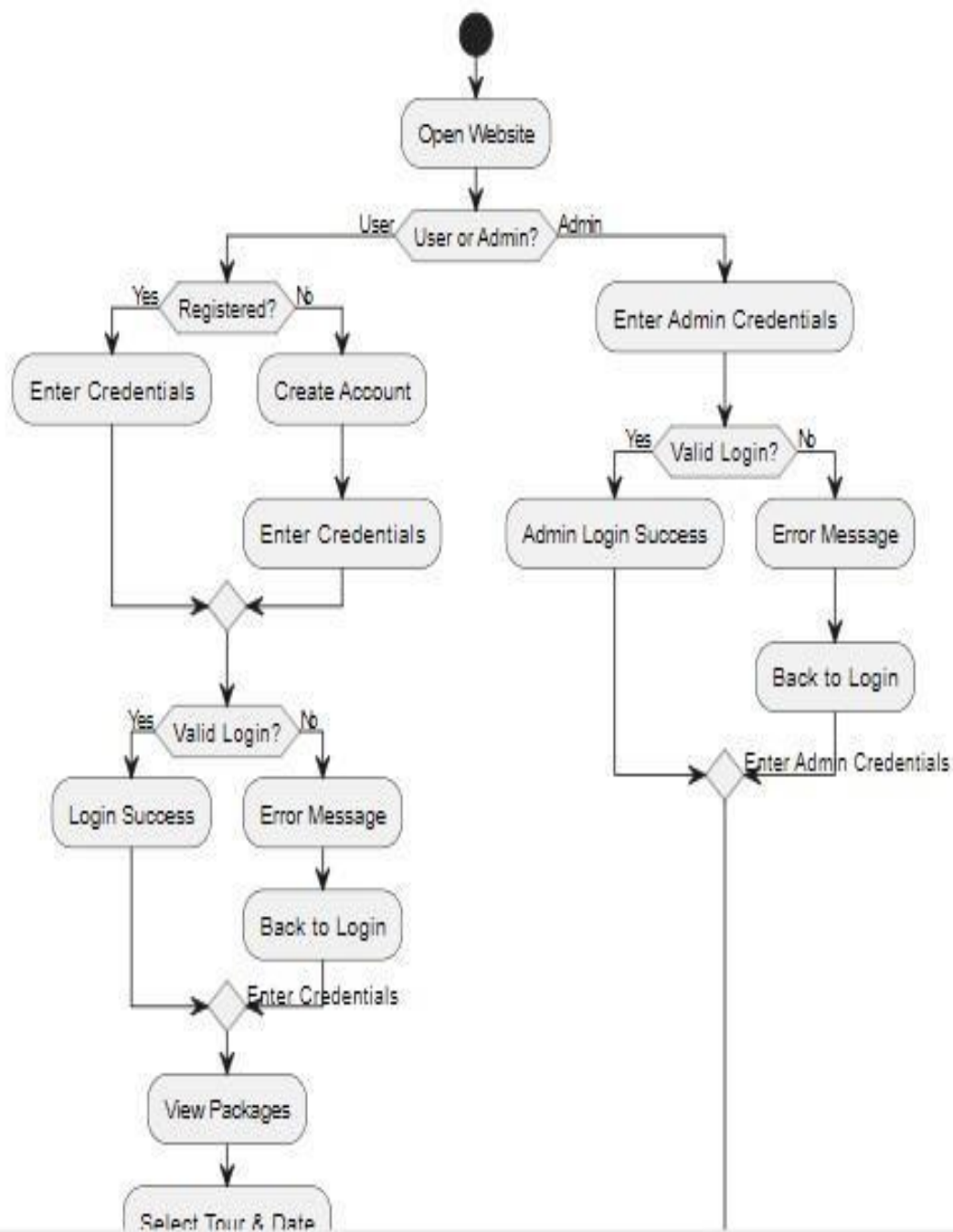
Architecture Diagram:

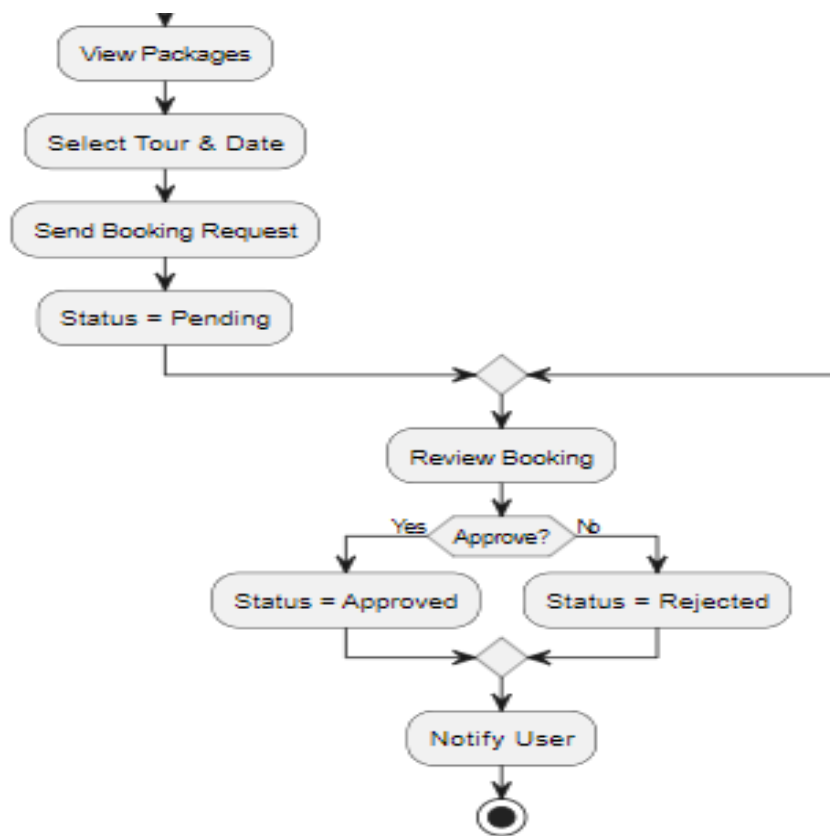


Activity Diagram:

An activity diagram represents the flow of activities performed in a system from start to end. It helps in understanding the decision making and workflow of processes such as tour booking and approval.

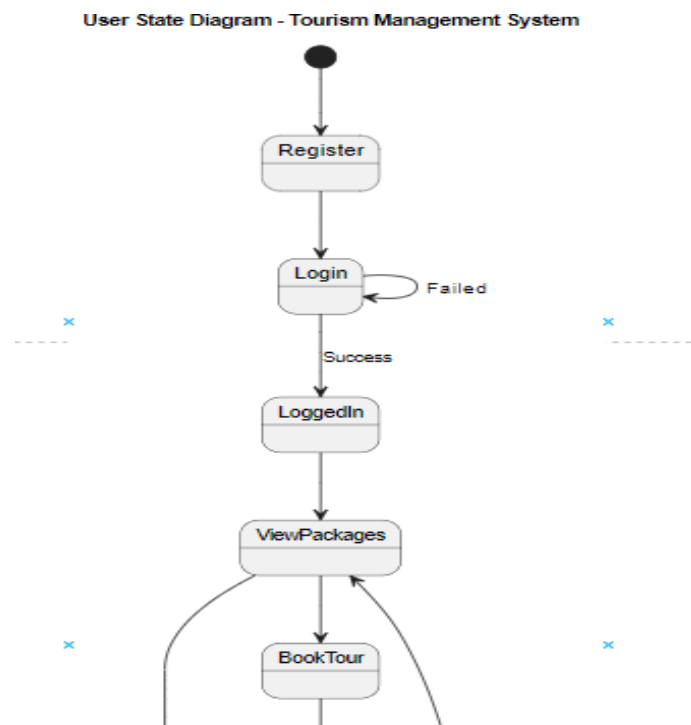
Activity Diagram - Tourism Management System

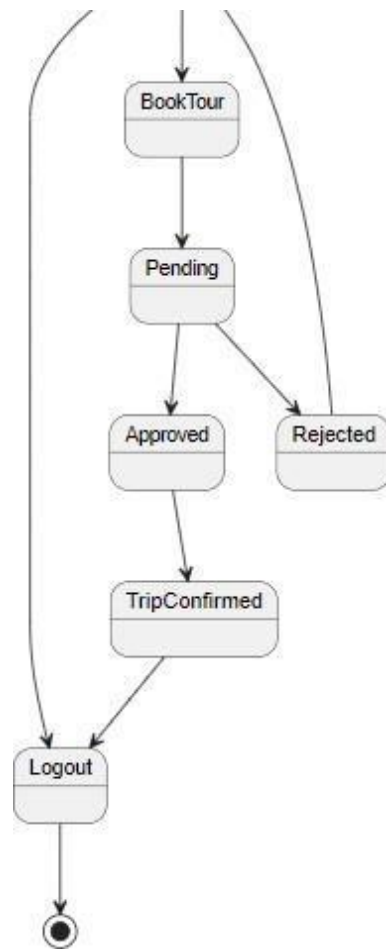




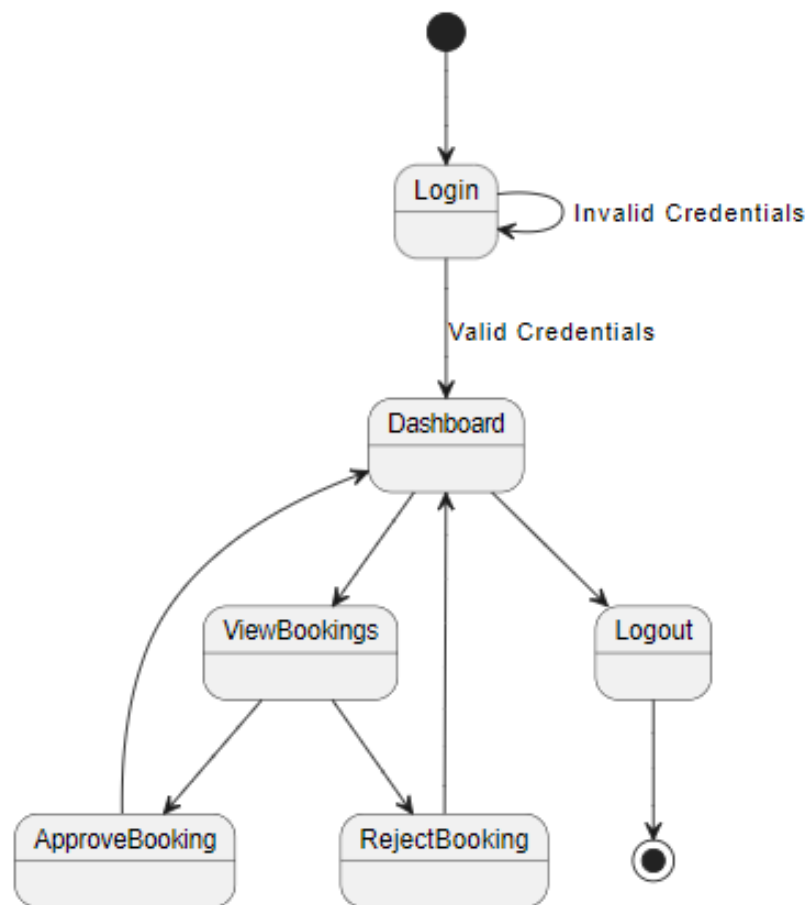
State diagram

A state diagram shows different states of a system and the transitions between them. It helps in understanding how the system behaves during various stages such as login, booking and approval.



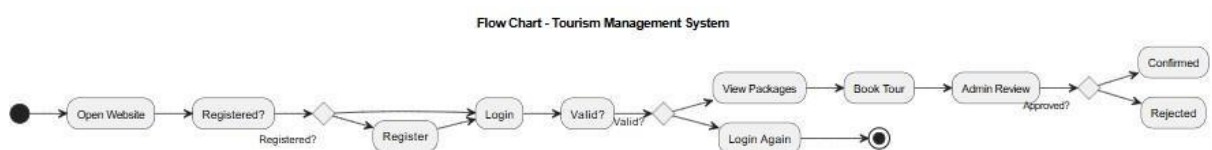


Admin State Diagram - Tourism Management System



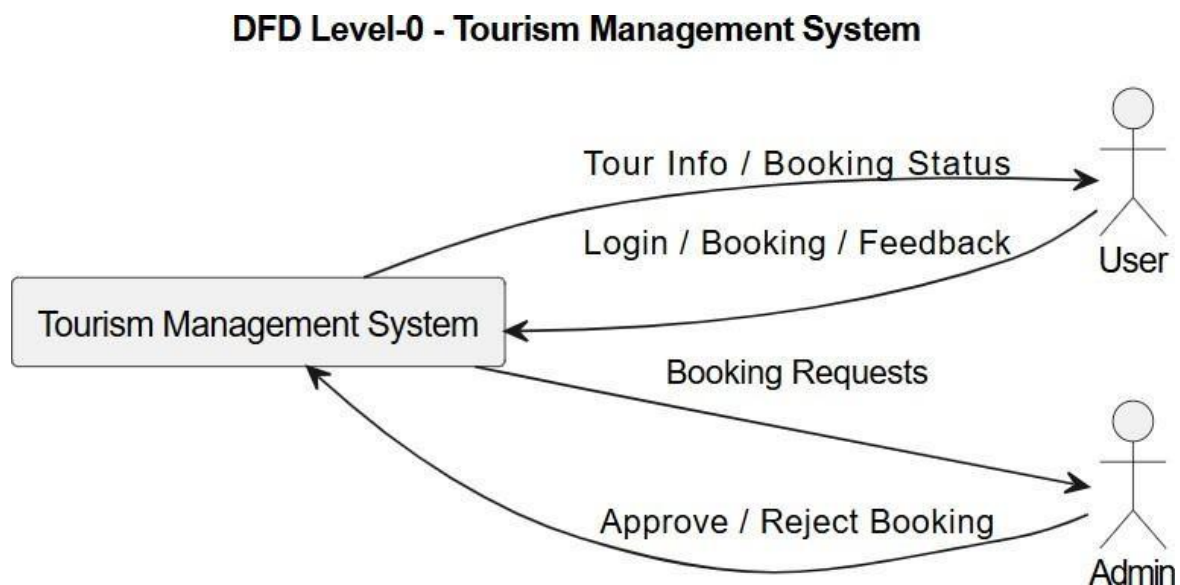
Flow Chart:

A flow chart is a graphical representation of the logical flow of a system. It helps in understanding the sequence of operations involved in tour booking and approval.



DFD:

This DFD Level-0 shows the interaction between users, admin and the Tourism Management System. It represents the overall data flow without internal process details

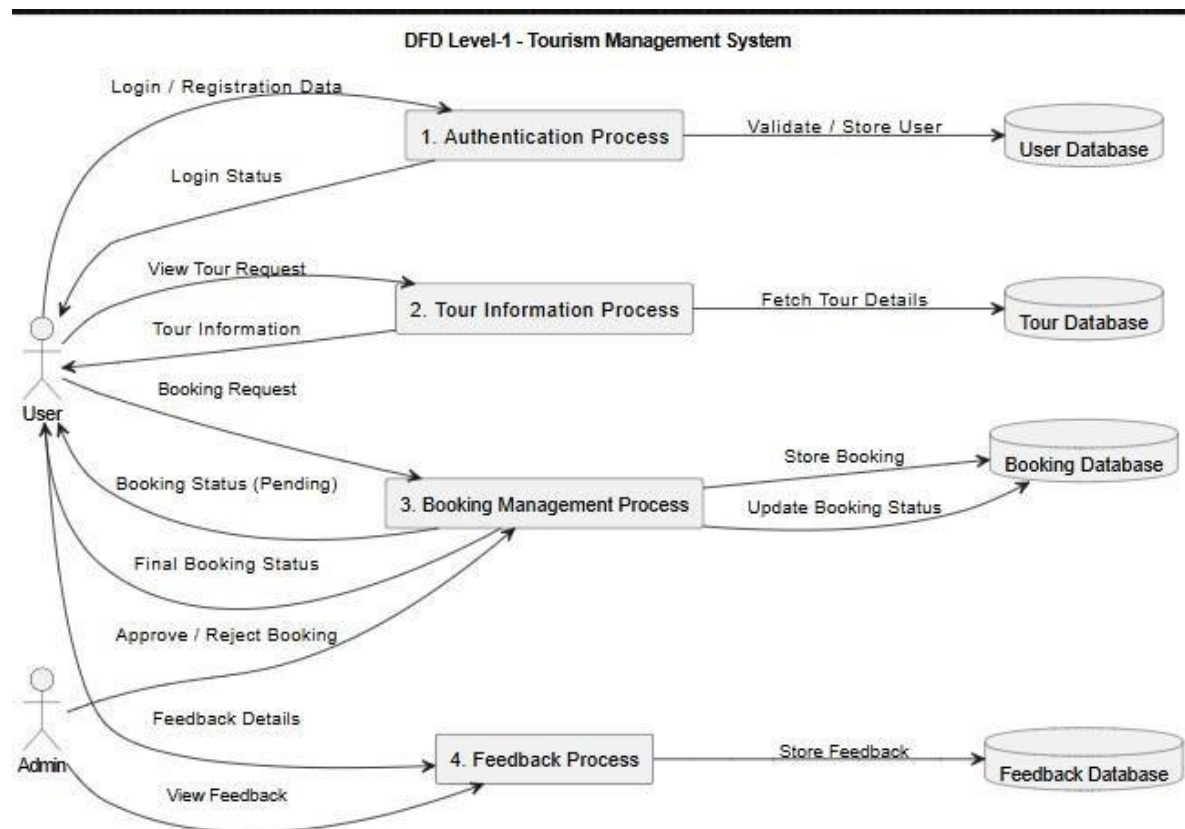


DFD Level-1

In this DFD Level-1, the Tourism Management System is divided into four major processes: Authentication, Tour Management, Booking Management and Feedback Management.

Users interact with the system to log in, view tours, book tours and submit feedback.

Admin verifies bookings and manages tour-related information. All data is stored and retrieved from relevant databases.

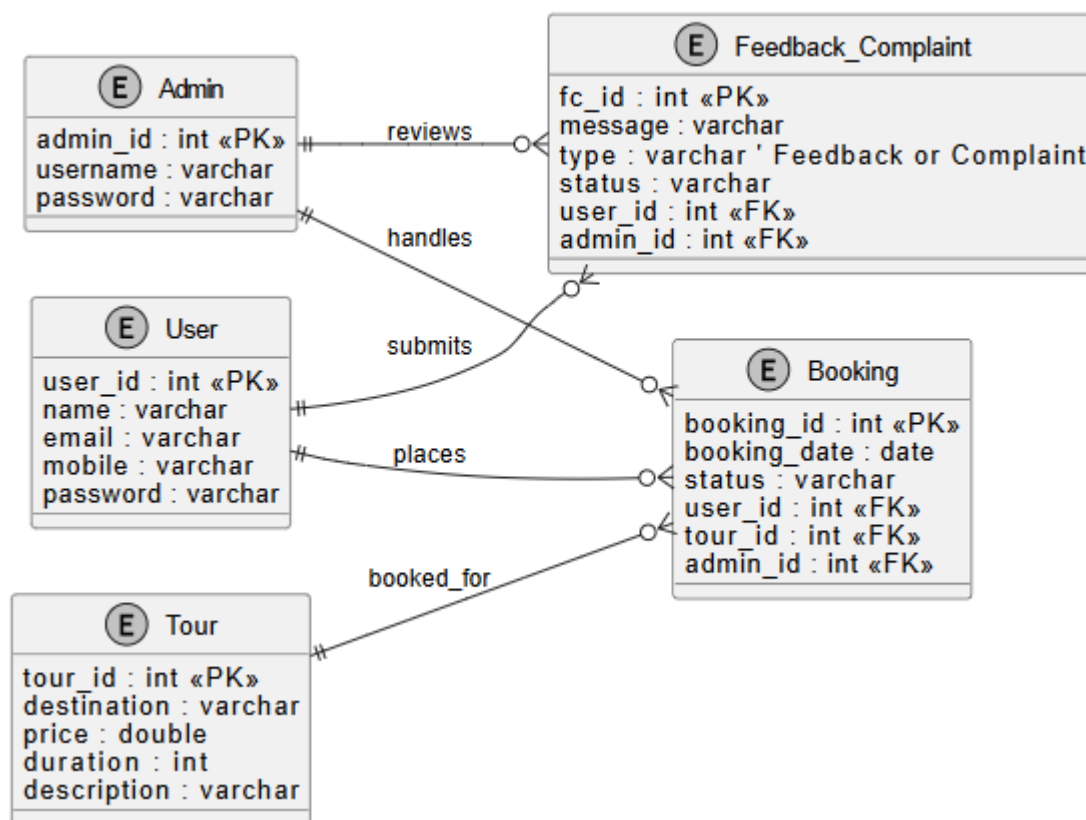


ER Diagram:

The ER Diagram of the Tourism Management System consists of entities such as User, Admin, Tour Package, Booking and Feedback.

Users can book tour packages, admins manage bookings and tours and users can submit feedback. Relationships between entities define how data is connected and stored in the database.

Entity Relationship Diagram - Tourism Management System



Description of Entities

The Tourism Management System database is designed to store and manage information related to users, tours, bookings, admin activities and feedback or complaints. Each entity represents an important part of the system and helps in maintaining data in an organized manner.

User Entity

The User entity stores information about customers who visit the tourism website. Users can register on the website, log in, view tour packages, book tours and submit feedback or complaints. Each user is uniquely identified using a user ID. The user details such as name, email, mobile number and password are stored for authentication and communication purposes.

Admin Entity

The admin entity contains information about administrators who manage the system. Admins log in using their credentials and are responsible for managing tour bookings and handling feedback or complaints submitted by users. Each admin is identified by a unique admin ID.

Tour Entity

The Tour entity stores details of all tour packages available on the website. It includes information such as destination, price, duration and description of the tour. Users can view these tour details and select a suitable tour package for booking.

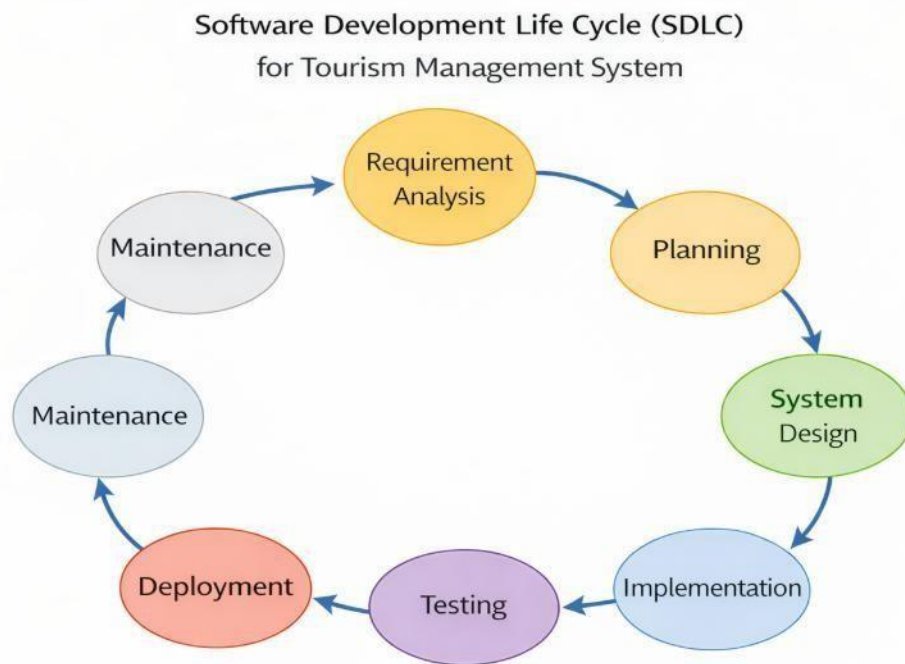
Booking Entity

The Booking entity stores information about tour bookings made by users. It keeps track of the booking date, booking status and the related user and tour details. This entity also records which admin handled the booking. Booking status helps users know whether their tour request is approved or rejected.

Feedback/Complaint Entity

The Feedback Complaint entity stores feedback and complaints submitted by users. Both are stored in a single table and differentiated using a type field. This entity also tracks the current status of the feedback or complaint and records which admin reviewed or handled it.

Software Development Life Cycle (SDLC):



Chapter 4

4. Testing and Validation

Testing and validation are important phases of software development that ensure the system works correctly and meets user requirements. In this project, testing was performed to verify that each module of the Tourism Management System functions as expected and produces accurate results.

4.1 Code Efficiency

Code efficiency refers to how well the system performs using minimum resources such as time and memory. In the Tourism Management System, efficient coding practices were followed to reduce unnecessary computations and improve performance.

Proper database queries were used to fetch only required data. Input validation was implemented to avoid incorrect data processing. Reusable functions and modular coding were used to improve maintainability and reduce redundancy in the code.

4.2 Testing Approach

A systematic testing approach was adopted to identify and fix errors at different stages of development. Testing was carried out during and after the implementation of each module.

Both **manual testing** and **functional testing** techniques were used. Test cases were designed based on user requirements to ensure that all features such as login, booking, approval, cancellation and feedback submission work correctly under different conditions.

4.3 Unit Testing

Unit testing focuses on testing individual modules or components of the system independently. In this project, each module was tested separately to verify its functionality.

Modules such as User Registration, User Login, Tour Booking, Admin Approval and Feedback Submission were tested individually using valid and invalid inputs. This helped in detecting errors at an early stage and ensured that each unit performed its intended task correctly.

Test Case ID	Module Name	Test Description	Input Data	Expected Result	Actual Result	Status
UT-01	User Registration	Verify new user registration	Valid user details	User registered successfully	As expected ,	Pass
UT-02	User Registration	Verify duplicate email registration	Existing email	Error message shown	As expected ,	Pass
UT-03	User Login	Verify login with valid credentials	Valid email & password	Login successful	As expected ,	Pass
UT-04	User Login	Verify login with invalid credentials	Invalid password	Error message displayed	As expected ,	Pass
UT-05	View Tours	Verify tour package display	Request tour list	Tour packages displayed	As expected ,	Pass
UT-06	Tour Booking	Verify booking request	Valid tour & date	Booking status set to Pending	As expected ,	Pass
UT-07	Booking Cancellation	Verify booking cancellation	Cancel request	Booking cancelled successfully	As expected ,	Pass
UT-08	Admin Login	Verify admin login	Valid admin credentials	Admin dashboard displayed	As expected ,	Pass
UT-09	Booking Approval	Verify booking approval	Approve request	Booking status updated to Approved	As expected ,	Pass
UT-10	Feedback/Complaint	Verify feedback submission	Feedback message	Feedback stored successfully	As expected ,	Pass

4.4 Integration Testing

Integration testing is done to check whether different modules of the system work properly together. After testing individual modules, this testing ensures that data flows correctly from one module to another.

In the Tourism Management System, integration testing checks the interaction between the user module, admin module and database entities. It verifies that user bookings are visible to the admin, admin decisions are updated for users and feedback or complaints submitted by users are correctly handled by the admin.

Test Case ID	Integrated Modules	Test Description	Input Data	Expected Result	Actual Result	Status
IT-01	User Registration & Login	Verify login after registration	Valid user details	User logged in successfully	As expected,	Pass
IT-02	Login & View Tours	Verify tour list after login	Valid credentials	Tours displayed	As expected,	Pass
IT-03	View Tours & Booking	Verify tour booking	Valid tour & date	Booking request created	As expected,	Pass
IT-04	Booking & Admin Approval	Verify booking approval	Approve request	Booking status updated	As expected,	Pass
IT-05	Booking & Cancellation	Verify booking cancellation	Cancel request	Booking cancelled	As expected,	Pass
IT-06	User & Feedback Module	Verify feedback submission	Feedback text	Feedback stored	As expected,	Pass
IT-07	Feedback & Admin Module	Verify admin feedback view	Admin login	Feedback displayed	As expected,	Pass

4.5 Screen Shot

4.5.1 Home Page



4.5.2 User Registration Page

×

Create your account

CREATE ACCOUNT

By logging in you agree to our [Terms and Conditions](#) and [Privacy Policy](#)

4.5.3 User Login Page

×

Signin with your account

[Forgot password](#)

SIGNIN

By logging in you agree to our [Terms and Conditions](#) and [Privacy Policy](#)

4.5.4 Forgot Password Page

Recover Password

Email id

Mobile No

New Password

Confirm Password

4.5.5 Tour Package List Page

Admin Login

Toll Number : 123-4568790 Sign Up / Sign In

Tourism Management System

SAFE & SECURE

HomeAboutTour PackagesPrivacy PolicyTerms of UseContact UsEnquiry

TMS- Package List

Package List



Package Name: Swiss Paris Delight Premium 2020 (Group Package)

Package Type : Group Package

Package Location : Paris and Switzerland

Features Round trip Economy class airfare valid for the duration of the holiday - Airport taxes - Accommodation for 3 nights in Paris and 3 nights in scenic Switzerland - Enjoy continental breakfasts every morning - Enjoy 5 Indian dinners in Mainland Europe - Exp

Rs 60000

[Details](#)



Package Name: Bhutan Holidays - Thimphu and Paro Special

Package Type : Family Package

Package Location : Bhutan

Features Free Wi-fi, Free Breakfast, Free Pickup and drop facility

Rs 30000

[Details](#)

4.5.6 Tour Booking Page



Swiss Paris Delight Premium 2020 (Group Package)

#PKG-1

Package Type : Group Package

Package Location : Paris and Switzerland

Features Round trip Economy class airfare valid for the duration of the holiday - Airport taxes - Accommodation for 3 nights in Paris and 3 nights in scenic Switzerland - Enjoy continental breakfasts every morning - Enjoy 5 Indian dinners in Mainland Europe - Exp

From

01-12-2025

To

05-12-2025

Grand Total
Rs. 60000

Package Details

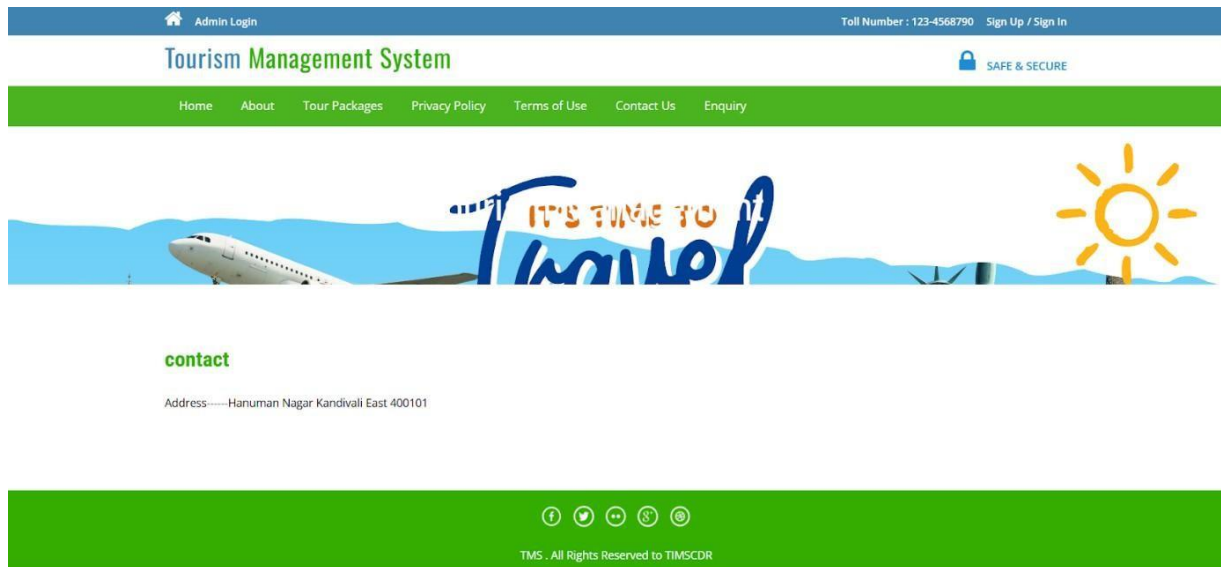
Pick this holiday for a relaxing vacation in Paris and Switzerland. Your tour embarks from Paris. Enjoy an excursion to popular attractions like the iconic Eiffel Tower. After experiencing the beautiful city, you will drive past mustard fields through Burgundy to reach Switzerland. While there, you can opt for a tour to Interlaken and then to the Trummelbach Falls. Photostop at Zurich Lake and a cable car ride to Mt. Titlis are the main highlights of the holiday.

Travels

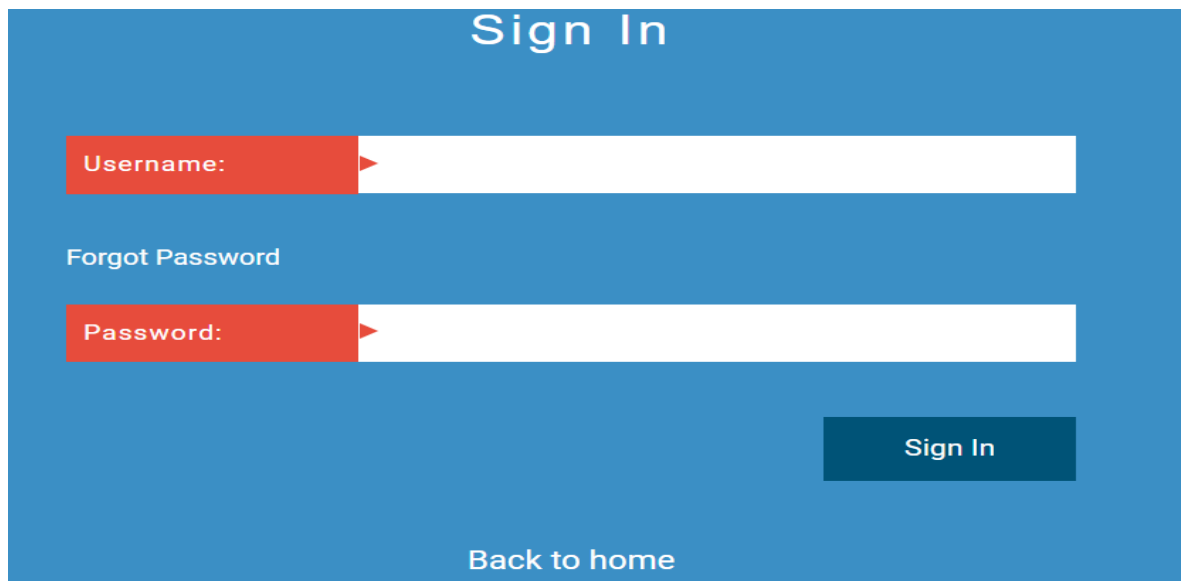
Comment

Book

4.5.7 Enquiry / Contact Page

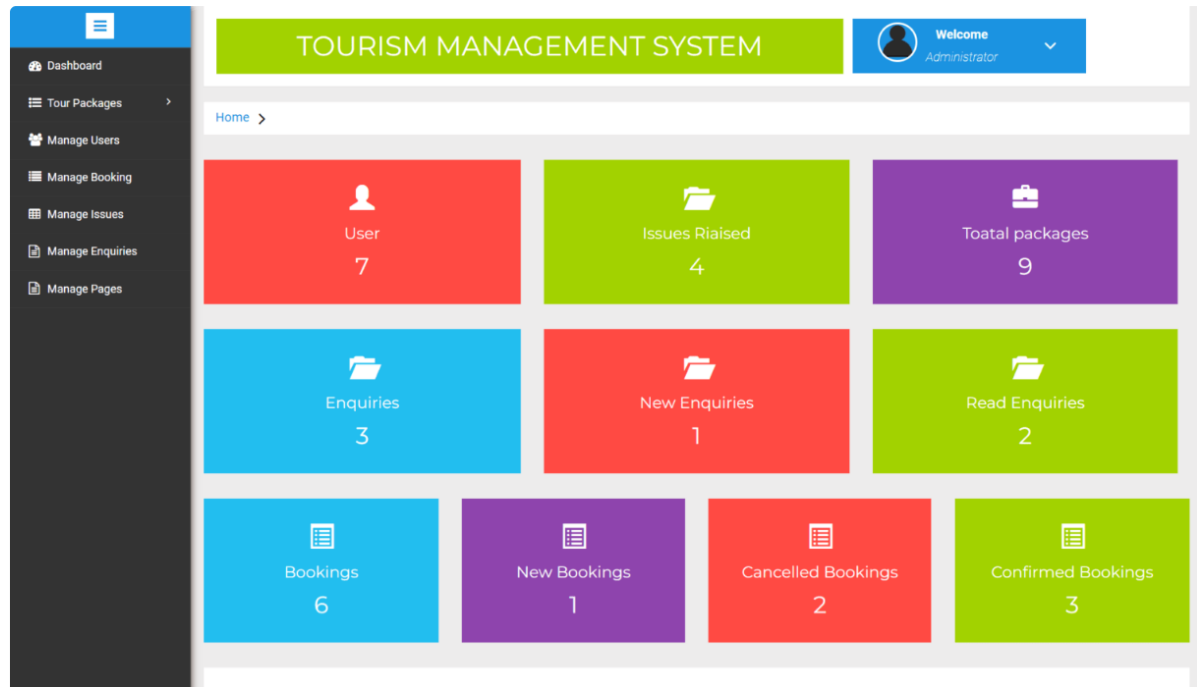


4.5.8 Admin Login Page



4.5.9 Admin Dashboard

Admin Dashboard



4.5.10 Manage / Create Package Page

Create Package

The Create Package page includes a sidebar with navigation links: Dashboard, Tour Packages, Manage Users, Manage Booking, Manage Issues, Manage Enquiries, and Manage Pages. The main content area contains a form with the following fields:

- Package Name: Create Package
- Package Type: Package Type eg: Family Package / Couple Package
- Package Location: Package Location
- Package Price in USD: Package Price is USD
- Package Features: Package Features Eg: free Pickup-drop facility
- Package Details: Package Details
- Package Image: Choose File | No file chosen

Buttons: CREATE, RESET

TMS. All Rights Reserved | TMS

Chapter 5

Implementation

4.1 Algorithm / Flowchart

This chapter explains the **core algorithms used in the Tourism Management System** for implementing user authentication, tour booking, enquiry handling and admin management.

Algorithm 1: User Registration Algorithm

Description

This algorithm allows a new user to register into the Tourism Management System by entering personal details. The system validates the input and stores user data securely in the database.

Algorithm Steps

1. Start
2. Display user registration form
3. User enters name, email, mobile number and password
4. Validate input fields
5. Check whether email already exists
6. If email exists, display error message
7. Else, encrypt password
8. Store user details in database
9. Display registration success message
10. End

Pseudo Code

START

INPUT name, email, mobile, password

IF email exists in database, THEN

 DISPLAY "Email already registered"

ELSE

 ENCRYPT password

 INSERT user details into database

 DISPLAY "Registration successful"

END IF

END

Algorithm 2: User Login Algorithm

Description

This algorithm validates user login credentials and grants access if they are correct.

Algorithm Steps

1. Start
2. Display login form
3. User enters email and password
4. Fetch user details from database
5. Compare encrypted password
6. If valid, allow login
7. Else, display error message
8. End

Pseudo Code

START

INPUT email, password

FETCH user record from database

IF password matches THEN

 CREATE session

 REDIRECT to user dashboard

ELSE

 DISPLAY "Invalid login credentials"

END IF

END

Algorithm 3: Tour Booking Algorithm

Description

This algorithm enables users to book a tour package by selecting a tour and preferred date.

Algorithm Steps

1. Start
2. User logs in
3. Display available tour packages
4. User selects tour package
5. User selects travel date

6. Store booking request
7. Set booking status as “Pending”
8. Display confirmation message
9. End

Pseudo Code

START

USER selects tour package and date

CREATE booking request

SET booking status = "Pending"

STORE booking details in database

DISPLAY "Booking request submitted"

END

Algorithm 4: Admin Booking Approval Algorithm

Description

This algorithm allows the admin to approve or reject booking requests.

Algorithm Steps

1. Start
2. Admin logs in
3. View pending bookings
4. Select booking request
5. Approve or reject booking
6. Update booking status
7. Notify user
8. End

Pseudo Code

START

ADMIN views pending bookings

SELECT booking

IF approve THEN

 UPDATE booking status = "Approved"

```
ELSE  
    UPDATE booking_status = "Rejected"  
END IF  
END
```

Algorithm 5: Enquiry and Feedback Algorithm

Description

This algorithm allows users to submit enquiries or feedback related to tours.

Algorithm Steps

1. Start
2. User enters enquiry details
3. Validate input
4. Store enquiry in database
5. Admin views enquiry
6. End

Pseudo Code

```
START  
INPUT enquiry details  
VALIDATE inputs  
STORE enquiry in database  
DISPLAY "Enquiry submitted successfully"  
END
```

Chapter 6

Cost and Benefit Analysis

Cost Analysis Using COCOMO II Model

In this project, the COCOMO II model is used to estimate the effort, development time and cost required for the Tourism Management System, which is a web-based application developed using PHP, MySQL, HTML, CSS and JavaScript. The project is developed by two team members and completed within a duration of 3 months.

COCOMO II Model Overview

COCOMO II estimates:

- **Effort** in Person-Months (PM)
- **Development Time (TDEV)** in months
- **Overall Cost**

The estimation is based on:

- Project size (KLOC)
- System complexity
- Team capability
- Tool usage

Project Size Estimation (KLOC)

The estimated size of the Tourism Management System is **12,000 Lines of Code (LOC)** i.e., **12 KLOC**.

Module-wise Code Distribution

	A	B	
1	Module-Wise Code Distribution		
2	Module	Technology	Estimated KLOC
3	Frontend	HTML,CSS, JavaScript	5 KLOC
4	Backend Logic	PHP	4 KLOC
5	Database & Integration	MySQL	3 KLOC
6		12 KLOC	12 KLOC
7			

COCOMO II Effort Estimation

COCOMO II Formula

$$\text{Effort (PM)} = A \times (\text{Size})^B \times \prod E_i$$

Where:

- **A = 2.94**
- **Size = 12 KLOC**
- **B = 1.10** (moderate complexity web application)
- **E_i = Effort Multipliers**

Effort Multipliers (E_i)

	A		B	C
1	Effort Multipliers			
2	Cost Driver	Description	Multiplier	Multiplier
1	RELY	Moderate reliability required		1.10
2	DATA	Medium database size		1.05
3	CPLX	Moderate system complexity		1.15
4	TEAM	Average team experience		1.10
5	TOOL	Standard development tools		1.00
6				

Combined Effort Multiplier

$$\text{Effort} = 1.10 \times 1.05 \times 1.15 \times 1.10 \times 1.00 \approx 1.46$$

Effort Calculation (Person-Months)

$$\text{Effort} = 2.94 \times (12)^{1.10} \times 1.46$$

$$\text{Effort} = 2.94 \times 15.45 \times 1.46$$

$$\text{Effort} \approx 66.8 \text{ Person-Months}$$

Total Estimated Effort = 66.8 Person-Months

Development Time (TDEV) Calculation

Formula

$$TDEV = C \times (\text{Effort})^D$$

Where:

- **C = 3.67**
- **D = 0.28**

$$TDEV = 3.67 \times (66.8)^{0.28} \approx 3 \text{ months}$$

Project Duration = 3 Months

Cost Estimation

Developer Cost Assumption

- Monthly cost per developer = **₹25,000**
- Team size = **2 developers**

Total Cost Calculation

$$\text{Total Cost} = \text{Effort (PM)} \times \text{Cost per PM}$$

$$\text{Total Cost} = 66.8 \times ₹7,500 \approx ₹5,01,000$$

Cost Breakdown Table

	A	B
1	Cost Component	Estimated Cost (₹)
2	Frontend Development	1,80,000
3	Backend Development	1,90,000
4	Database Design & Integration	80,000
5	Testing & Deployment	50,000
6	Total Cost	₹5,00,000
7		

Effort Distribution (3-Month Plan)

	A	B	C
1	Phase	Percentage	Effort (PM)
2	Planning & Requirement Analysis	15%	10.0
3	System Design	20%	13.4
4	Implementation	40%	26.7
5	Testing	15%	10.0
6	Deployment & Documentation	10%	6.7
7	Total	100%	₹6,8 PM
8			

Benefits of Tourism Management System

1. Operational Efficiency

- Automates tour booking and management
- Reduces manual paperwork

2. Time Saving

- Users can book tours online
- Admin can manage packages efficiently

3. Cost Reduction

- Minimizes operational and manpower costs
- No licensing cost (open-source tools)

4. Accessibility

- System is available 24×7
- Accessible from any browser-enabled device

5. Business Growth

- Improves customer experience
- Enables future enhancements such as online payment and mobile app integration

Risk Analysis

Technical Risks

Technical Risks		
Risk	Description	Mitigation
Server Failure	Hosting issues	Regular backups
Security Issues	Unauthorized access	Authentication & validation

Operational Risks

Operational Risks		
Risk	Description	Mitigation
Scope Change	Additional features	Fixed scope planning
Time Overrun	Delay in coding	Module-wise deadlines
Total		₹6.8 PM

Summary Table

	A	B
1	Summary Table	
2	Parameter	Value
3	Project Duration	3 Months
4	Team Size	2 Developers
5	Total Effort	66.8 Person-Months
6	Estimated Cost	₹5,00,000
7	Development Model	COCOMO II
8		

Conclusion

Using the **COCOMO II model**, the Tourism Management System is estimated to require **66.8 person-months**, with a **development duration of 3 months** and a total cost of approximately **₹5,00,000**.

The benefits gained from automation, efficiency and scalability outweigh the development cost, making the system **cost-effective and practical**.

Software Parameter Estimation

1. Effort Estimation (Function Point Analysis – FPA)

Function Point Analysis (FPA) is used to estimate the software size and effort based on system functionality from the user's perspective. The estimated function points for the **Tourism Management System** are as follows:

Function Point Components

- **Inputs:**

User input forms such as user registration, login, booking form, enquiry form, feedback form.

Estimated FP: 10

- **Outputs:**

Display of tour packages, booking confirmation, booking status, admin reports.

Estimated FP: 8

- **User Queries:**

Search tour packages, view booking history, check booking status.

Estimated FP: 6

- **Logical File:**

Database files containing user details, tour packages, bookings, enquiries and feedback.

Estimated FP: 12

- **Interfaces:**

Interaction between frontend, backend and database modules.

Estimated FP: 4

Total Function Points

Total Function Points (FP) = $10 + 8 + 6 + 12 + 4 = 40$ FPs

2. Effort Calculation (in Person-Days)

For a web-based system of moderate complexity, productivity is assumed as:

- **1 Function Point = 1.75 Person-Days**

Effort Calculation

Effort = Function Points × Person-Days per FP

Effort = 40×1.75

Effort = 70 Person-Days

Total Effort Required = 70 Person-Days

3. Time and Schedule Estimation

Team Size

- **2Developers:**

(Handling frontend, backend, database integration and testing)

Project Duration

- Total project duration = **3 Months**
- Working days per month $\approx$ **24 days**
- Total available working days $\approx$ **72 days**

Effort vs Duration

- Required effort = **70 days**
- Available duration = **72 days**

The project comfortably fits within the **3-month duration**.

4. Effort Distribution

Effort Distribution		
Phase	Effort Distribution	Person-Days
Requirement Analysis & Planning	15%	11 days
System Design	20%	14 days
Implementation	40%	28 days
Testing & Integration	15%	11 days
Deployment & Documentation	10%	6 days
Total	100%	70 days

5. Risk Analysis

a. Technical Risks

Server Failure:

Risk: Hosting or server downtime

Mitigation: Regular backups and stable hosting environment

Security Issues:

Risk: Unauthorized access to user data

Mitigation: Authentication, session management and input validation

b. Operational Risks

Scope Change:

Risk: Addition of new features during development

Mitigation: Fixed scope planning and approval process

Time Overrun:

Risk: Delay in module implementation

Mitigation: Module-wise deadlines and regular progress review.

Summary Based on FPA

Summary Based on FPA	
Parameter	Value
Project Type	Web-based Tourism Management System
Estimation Technique	Function Point Analysis
Total Function Points	40
Total Effort	70 Person-Days
Team Size	2 Developers
Project Duration	3 Months
Risk Type	Technical & Operational

Conclusion

Using **Function Point Analysis**, the Tourism Management System is estimated to require **70 person-days** of effort. With **two developers working over a period of 3 months**, the project can be completed efficiently within the planned schedule. The estimation confirms that the project is **feasible, well-planned and cost-effective**, with manageable risks and clear effort distribution.

Chapter 7

7. User Manual

The User Manual provides clear instructions on how to use the Tourism Management System. It explains the steps required by users and admins to operate the system smoothly.

6.1 User Documentation

6.1.1 User (Tourist) Guide

1. Open the Tourism Management System website using a web browser.
2. If you are a new user, click on the **Register** option and enter required details.
3. Login using your registered email and password.
4. After login, view the list of available tour packages.
5. Select a tour package and choose the preferred travel date.
6. Submit the booking request.
7. Check booking status (Pending / Approved / Rejected).
8. If required, cancel the booking.
9. Submit feedback or raise a complaint after the tour.
10. Logout from the system after use.

6.1.2 Admin Guide

1. Login using admin credentials.
2. View all booking requests submitted by users.
3. Approve or reject tour bookings.
4. Manage tour package details if required.
5. View and handle feedback or complaints raised by users.
6. Logout securely after completing tasks.

Chapter 8

8. Conclusion

7.1 Conclusion

The Tourism Management System is a web-based application developed to simplify the process of viewing and booking tour packages. The system provides an easy interface for users to register, log in, explore tours and book trips online. It also allows the admin to manage tour packages, approve or reject bookings and handle feedback or complaints.

This project successfully reduces manual work and improves communication between users and administrators. Overall, the system meets the project objectives and provides a reliable solution for managing tourism services efficiently.

7.2 Limitations

Although the Tourism Management System works effectively, it has some limitations. Online payment functionality is not included in the current version, so bookings are confirmed only after admin approval. The system depends completely on internet connectivity to function properly.

The application does not provide real-time notifications for booking updates. It also lacks mobile application support, which limits accessibility for users who prefer smartphones. Advanced features such as dynamic pricing, live availability of tours and GPS-based location services are not included. The system also does not integrate with government or tourism department portals.

7.3 Future Scope

The Tourism Management System can be enhanced further by adding online payment options for instant booking confirmation. A mobile application can be developed to improve accessibility and user convenience.

In the future, the system can be integrated with tourism platforms similar to the **Incredible India** website to provide official tour information, cultural details and government-approved travel packages. Features such as tour recommendations based on user interest, real-time notifications and multilingual support can also be added.

Additional enhancements may include live chat support, rating and review systems, GPS-based tourist guidance, hotel and transport booking integration and seasonal offers. These improvements will make the system more advanced, user-friendly and suitable for large-scale tourism management

Chapter 9

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